```
Name     RAMSEL;
PartNo   00;
Date     13/01/2026;
Revision 01;
Designer F. Spencer;
Company  Design;
Assembly None;
Location UAE;
Device   g16v8a;

/* THIS IS CODE FOR DUBAI-Clone-XT created by Felipe Spencer Picada*/

/*******************************************************************************

- PC XT RAM Address Decoder
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- This decoder manages 1MB (1024KB) of addressable memory space in a PC XT
- system using address lines A0-A19 (20 address lines = 2^20 = 1MB).
-
- MEMORY MAP OVERVIEW (PC XT Standard):
- ========================================
- 0x00000-0x9FFFF (0-639KB)   : Conventional RAM (640KB)
- 0xA0000-0xBFFFF (640-767KB) : Video RAM (128KB)
- 0xC0000-0xFFFFF (768-1023KB): ROM & Adapters (256KB)
-
- ADDRESS LINE USAGE:
- =====================
- A0-A15  : Offset within 64KB segment (handles 0-65535)
- A16-A19 : Segment selector (defines which 64KB block)
-
- SEGMENT DECODING (Based on A16-A19):
- ========================================
- A19 A18 A17 A16   | Binary Range          | Hex Range          | Segment | Size
- 0   0   0   0     | 00000-0FFFF (64KB)  | 0x00000-0x0FFFF| 0        | 64KB
- 0   0   0   1     | 10000-1FFFF (64KB)  | 0x10000-0x1FFFF| 1        | 64KB
- 0   0   1   0     | 20000-2FFFF (64KB)  | 0x20000-0x2FFFF| 2        | 64KB
- 0   0   1   1     | 30000-3FFFF (64KB)  | 0x30000-0x3FFFF| 3        | 64KB
- 0   1   0   0     | 40000-4FFFF (64KB)  | 0x40000-0x4FFFF| 4        | 64KB
- 0   1   0   1     | 50000-5FFFF (64KB)  | 0x50000-0x5FFFF| 5        | 64KB
- 0   1   1   0     | 60000-6FFFF (64KB)  | 0x60000-0x6FFFF| 6        | 64KB
- 0   1   1   1     | 70000-7FFFF (64KB)  | 0x70000-0x7FFFF| 7        | 64KB
- 1   0   0   0     | 80000-8FFFF (64KB)  | 0x80000-0x8FFFF| 8        | 64KB
- 1   0   0   1     | 90000-9FFFF (64KB)  | 0x90000-0x9FFFF| 9        | 64KB
- 1   0   1   0     | A0000-AFFFF (64KB)  | 0xA0000-0xAFFFF| A(VGA)  | 64KB
- 1   0   1   1     | B0000-BFFFF (64KB)  | 0xB0000-0xBFFFF| B(VGA)  | 64KB
- 1   1   0   0     | C0000-CFFFF (64KB)  | 0xC0000-0xCFFFF|C(ROM)  | 64KB
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```
- 1   1   0   1          | D0000-DFFFF (64KB) | 0xD0000-0xDFFFF| D(ROM) | 64KB
- 1   1   1   0          | E0000-EFFFF (64KB)  | 0xE0000-0xEFFFF| E(ROM)  | 64KB
- 1   1   1   1          | F0000-FFFFF (64KB)  | 0xF0000-0xFFFFF| F(BIOS)   | 64KB

*******************************************************************************/
/* Inputs - Address Lines */

PIN 1  = A16;  /* USED for eprom - FUTURE */
PIN 2  = A17;
PIN 3  = A18;
PIN 4  = A19;
PIN 5  = PLANARRAM0;  /* Future */
PIN 6  = PLANARRAM1;  /* Future */
PIN 7  = A15;  /* USED for eprom - FUTURE */

/* Note: A0-A15 handle offset within each 64KB segment (not used in this decoder) */

/* Outputs - RAM Control Signals */

PIN 12 = RAMADDR;  /* RAM Address Valid - Active HIGH when RAM region
accessed */

PIN 13 = !CSSTATIC128; /* U102 -  Active Low CE128K0 618128 */
PIN 15 = !CSSTATIC; /* U99 - Active Low CE512K0 AS6C4008 */

/* Future - JUMPERS: F1, F2, F3 */

/* U101 OPPTIONAL EXTRA 512K CE512K1- Active Low AS6C4008 */
/* PIN 14 = !CSSTATICEXTRA; NOT USED */

/* U18A CS7 OPPTIONAL ROM 27C64 */
/* PIN 18 = !BIOSROMCS7; NOT USED */

/* U19A CS6 OPPTIONAL ROM 27C64 */
/* PIN 17 = !BIOSROMCS6; NOT USED */

/* EPROM ROM LATCH SIGNAL FOR U23*/
/* PIN 19 = ROMADDR; NOT USED */

/*******************************************************************************
  Equations
*******************************************************************************/

/* Logic Equations */

/* RAMADDR - RAM Address Valid Signal
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```
/* STATIC ONLY 512K PLUS 128K TWO CHIPS*/
RAMADDR = !A19 # A19 & !A18 & !A17;
CSSTATIC = !A19;
CSSTATIC128 = A19 & !A18 & !A17;
```